RIPHAH INTERNATIONAL UNIVERSITY ISLAMABAD

Faculty of Social Sciences & Humanities

Department of Computer Science

Final Term Examination Spring 2026 Semester

Course Name:	Object-Oriented Programming	**Course Code:**	CS201
Program:	BS (Computer Science)	**Section:**	1
Semester:	Final	**Date of Examination**	01-06-2026
Teacher's Name	Muhammad Usman	**Total Marks**	40
Exam Duration	(10:00 am – 12:00 pm)		

INSTRUCTIONS

- Write your name and SAP ID on all answer sheets.
- Write page number on all sheets.
- Please be analytical and specific in answering questions.
- Switch off your Mobile Phone.
- Attempt all the following questions critically and creatively.

Question 1: **[12 Marks]**

Answer the following short scenario-based questions:

(a) A student claims that 'Encapsulation is just about making data private.' Do you agree or disagree? Justify your answer with a real-world analogy. [3 Marks]

(b) Explain the difference between Method Overloading and Method Overriding with the help of an example scenario for each. [3 Marks]

(c) A developer creates a class Animal with a method makeSound(). He creates subclasses Dog and Cat. Why should makeSound() be declared abstract in Animal? What happens if it is not declared abstract? [3 Marks]

(d) Differentiate between a class and an object using a real-world scenario. Why is a class called a blueprint? [3 Marks]

Question 2: **[08 Marks]**

(a) Identify and correct ALL errors in the following code:

```
class Rectangle {
    private int width;
    private int height;

    public Rectangle(int w, int h) {
        width = w;
        height = h;
    }

    public int getArea() {
```

```java
        return width * height
    }
}

class Main {
    public static void main(String args[]) {
        Rectangle r = new Rectangle(5, 10);
        System.out.println("Area: " + r.area());
    }
}
```

(b) Analyze the following code and write the exact output:

```java
class Counter {
    static int count = 0;
    Counter() { count++; }
    void display() { System.out.println("Count: " + count); }
}

class Main {
    public static void main(String[] args) {
        Counter c1 = new Counter();
        Counter c2 = new Counter();
        Counter c3 = new Counter();
        c1.display();
        c3.display();
    }
}
```

Question 3: **[20 Marks]**

Design a Library Management System in Java that demonstrates the following OOP concepts:

1. Inheritance: Create a base class LibraryItem and derive Book and Magazine from it.
2. Encapsulation: Use private fields with proper getters and setters.
3. Polymorphism: Override a displayInfo() method in each subclass.
4. Abstraction: Use an abstract class or interface for common item behaviour.
5. File Handling: Save and read borrowed item records to/from a text file.
6. Exception Handling: Handle item not found, file not found, and invalid input using try-catch blocks.

Note: Your program should allow adding items, borrowing an item (saved to file), and displaying all items.